



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2022 Year 12 Course Assessment Task 4 (Trial Examination)

Thursday 14 August, 2022

General instructions

- Working time – 3 hours.
(plus 10 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided (on page 13)

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: # BOOKLETS USED:

Class (please ✓)

☐ 12MXX.1 – Miss Ham

☐ 12MXX.2 – Mr Sekaran

Marker's use only.

QUESTION	1-10	11	12	13	14	15	16	%
MARKS	$\overline{10}$	$\overline{14}$	$\overline{12}$	$\overline{15}$	$\overline{17}$	$\overline{17}$	$\overline{15}$	$\overline{100}$

Section I

10 marks

Attempt Question 1 to 10

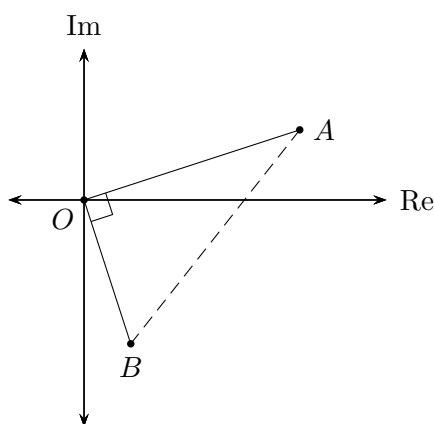
Allow approximately 15 minutes for this section

Mark your answers on the answer grid provided (labelled as page 13).

Questions

Marks

1. The Argand diagram shows $\triangle OAB$ where A represents the complex number $3e^{i\frac{\pi}{10}}$, $\angle BOA = 90^\circ$, and $|\overrightarrow{OB}| = \frac{2}{3}|\overrightarrow{OA}|$. 1



Which of the following represents the complex number B ?

- (A) $2e^{-i\frac{2\pi}{5}}$ (C) $2e^{-i\frac{9\pi}{10}}$
 (B) $2e^{i\frac{3\pi}{5}}$ (D) $3e^{i\frac{3\pi}{5}}$
2. Let $z = \cos \theta + i \sin \theta$. Which of the following is the expression for $\operatorname{Im}\left(\frac{1}{z^5} - \bar{z}^3\right)$? 1

- (A) $\cos 5\theta + \cos 3\theta$ (C) $-\sin 5\theta - \sin 3\theta$
 (B) $\cos 5\theta - \cos 3\theta$ (D) $\sin 3\theta - \sin 5\theta$

3. Consider the lines $\ell_1 : \underline{r}_1 = \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -3 \\ 4 \end{pmatrix}$ and $\ell_2 : \underline{r}_2 = \begin{pmatrix} 6 \\ -2 \\ -5 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -3 \\ -2 \end{pmatrix}$. 1

Which of the following points do ℓ_1 and ℓ_2 intersect at?

- (A) $(3, -5, 5)$ (C) $(7, 4, -7)$
 (B) $(5, 1, -3)$ (D) ℓ_1 and ℓ_2 are skew

4. A particle moves in a straight line and its motion is described by 1

$$v^2 = 6(7 - 12x - 2x^2)$$

where x is its displacement from the origin in metres and v is its velocity in ms^{-1} .

Which of the following statements is true?

(A) Frequency is 12 ; Centre of motion is $x = -3$

(B) Frequency is 24 ; Centre of motion is $x = 3$

(C) Frequency is $2\sqrt{3}$; Centre of motion is $x = -3$

(D) Frequency is $2\sqrt{6}$; Centre of motion is $x = 3$

5. The points P , Q and R are collinear where $\overrightarrow{OP} = \underline{\underline{i}} - \underline{\underline{j}}$, $\overrightarrow{OQ} = -3\underline{\underline{j}} - \underline{\underline{k}}$ and $\overrightarrow{OR} = 2\underline{\underline{i}} + m\underline{\underline{j}} + n\underline{\underline{k}}$ for some constants m and n . 1

Which of the following are possible values for m and n ?

(A) $m = -1$ and $n = -1$

(C) $m = 1$ and $n = 1$

(B) $m = -1$ and $n = 1$

(D) $m = 1$ and $n = -1$

6. Which of the following is the negation of $p \Rightarrow (q \wedge r)$? 1

(A) $\sim p \Rightarrow (q \wedge r)$

(C) $\sim p \wedge (q \wedge r)$

(B) $(\sim q \vee \sim r) \Rightarrow \sim p$

(D) $p \wedge (\sim q \vee \sim r)$

7. Which of the following expressions is equivalent to $\int_0^{\frac{\pi}{6}} e^{\frac{x}{2}} \sin 3x \, dx$? 1

(A) $-6 - 2 \int_0^{\frac{\pi}{6}} e^{\frac{x}{2}} \sin 3x \, dx$

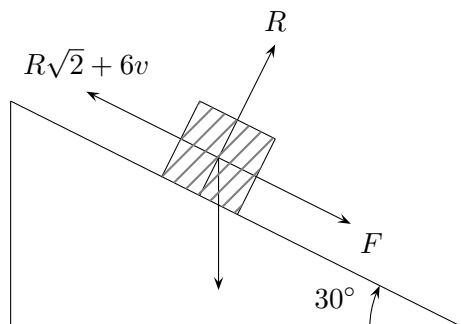
(C) $6 + 2 \int_0^{\frac{\pi}{6}} e^{\frac{x}{2}} \sin 3x \, dx$

(B) $2e^{\frac{\pi}{12}} - 6 \int_0^{\frac{\pi}{6}} e^{\frac{x}{2}} \cos 3x \, dx$

(D) $\frac{1}{2}e^{\frac{\pi}{12}} - \frac{3}{2} \int_0^{\frac{\pi}{6}} e^{\frac{x}{2}} \cos 3x \, dx$

8. An object of mass 50 kg is pulled by a constant force of F newtons, down a long rough slope inclined at 30° to the horizontal. 1

The object is met with a total resistive force of $R\sqrt{2} + 6v$, where R is the normal reaction force exerted by the slope on the object and v is the velocity of the object in ms^{-1} . The acceleration of the object along the slope is $a \text{ ms}^{-2}$ and the acceleration due to gravity is $g \text{ ms}^{-2}$.

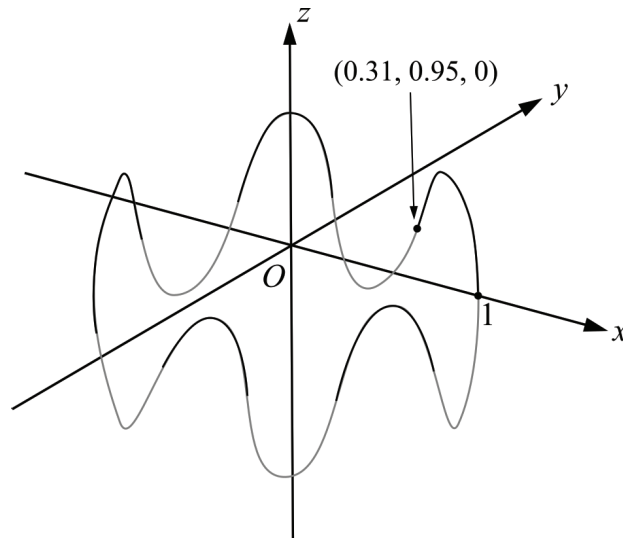


Which of the following is an expression for F ?

- (A) $F = 50a - 25g + R\sqrt{2} + 6v$ (C) $F = -R\sqrt{2} - 6v$
 (B) $F = 50a - 25\sqrt{3}g + R\sqrt{2} + 6v$ (D) $F = 25g + R\sqrt{2} + 6v$
9. Which of the following statements is FALSE? 1
- (A) $\exists x \in \mathbb{Z}, \forall y \in \mathbb{R}$ such that $\sqrt{x} < e^y$
 (B) $\exists x \in \mathbb{R}^+, \forall y \in \mathbb{R}$ such that $|y| < x$
 (C) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $\cos x = \cos(-y)$
 (D) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}^+$ such that $x^4 = 2y^3$

10. A curve in three dimensions is represented by vector $\underline{r}(t)$ as shown in the diagram below

1



Which of the following is a possible vector for $\underline{r}(t)$?

- (A) $\underline{r}(t) = \cos(t)\underline{i} + \sin(t)\underline{j} + \sin(5t)\underline{k}$
- (B) $\underline{r}(t) = \sin(5t)\underline{i} + \cos(t)\underline{j} + \sin(t)\underline{k}$
- (C) $\underline{r}(t) = \cos(5t)\underline{i} + \sin(t)\underline{j} + \cos(t)\underline{k}$
- (D) $\underline{r}(t) = \sin(t)\underline{i} + \cos(t)\underline{j} + \cos(5t)\underline{k}$

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (14 Marks)		Commence a NEW booklet.	Marks
(a)	i. Find the Cartesian equation defined by		2
	$\left z - 2(1 + i) \right = \left z + 6(1 + i) \right $		
	ii. Hence sketch the region on the Argand diagram defined by		1
	$\left z - 2(1 + i) \right \leq \left z + 6(1 + i) \right $		
(b)	Solve for w given that $w^3 = 32\sqrt{3} - 32i$.		3
(c)	Find α and β given that $z^3 + 9z + 6\sqrt{3}i = (z - \alpha)(z - \beta)^2$.		3
(d)	Consider the complex number $z = \tan \alpha + i$ where $0 < \alpha < \frac{\pi}{2}$.		
	i. Show that $z = \sec \alpha \times e^{(\frac{\pi}{2} - \alpha)i}$.		2
	ii. Hence find all values of α for which z^{-2} is purely imaginary.		3

Examination continues overleaf...

Question 12 (12 Marks)

Commence a NEW booklet.

Marks

- (a) By expressing as a sum of partial fractions, find

3

$$\int \frac{7x^2 - 3}{(x^2 + 6x - 3)(2x - 1)} dx$$

- (b) Use an appropriate substitution to find

4

$$\int \frac{3}{2 - \sin 2x} dx$$

- (c) i. Show that
- $\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx = \frac{\pi}{12} - \frac{\sqrt{3}}{8}$
- .

3

- ii. Hence, or otherwise, evaluate

2

$$\int_0^{\frac{1}{2}} x \cos^{-1} x dx$$

Examination continues overleaf...

Question 13 (15 Marks) Commence a NEW booklet. **Marks**

- (a) i. If $I_n = \int_1^e (1 - \ln x)^n dx$ for $n \in \mathbb{Z}^+$, show for $n \geq 1$ that **2**

$$I_n = -1 + nI_{n-1}$$

- ii. Use the principle of mathematical induction to prove that **4**

$$I_n = n!e - 1 - \sum_{r=1}^n \frac{n!}{(n-r)!}$$

for all positive integers $n \geq 1$.

- (b) A particle on a spring is moving horizontally in simple harmonic motion and x cm is its horizontal displacement from the origin. The particle is initially at its centre of motion and is moving to the right. Its acceleration is given by

$$\ddot{x} = -\frac{x}{16} - \frac{1}{4}$$

The particle has an amplitude of 8 cm and frequency n .

- i. State the maximum displacement of the particle. **1**
 ii. The particle has velocity v cms⁻¹. By integral methods, show that **3**

$$v^2 = \frac{1}{16} (64 - (x+4)^2)$$

- iii. Find the maximum speed of the particle. **1**

The displacement of the particle is given by $x = 8\sin(nt) + k$, where t is the time in seconds after it begins moving right from its centre of motion.

- iv. State the value of n and k . **1**
 v. Find the times when the magnitude of the particle's acceleration is the greatest for $0 \leq t \leq 8\pi$. **2**
 vi. Hence find when the particle has travelled a total distance of 40 cm. **1**

Question 14 (17 Marks)	Commence a NEW booklet.	Marks
(a) Prove that if $a, b \in \mathbb{R}$,	$\frac{5}{8}a^4 + \frac{2}{5}b^2 \geq a^2b$	2
(b) Prove that there is no complex number which satisfies the equation	$ 3z - 3z = -\frac{1}{2}i$	2
(c) i. By considering the expansion of $(x + y)^2$, or otherwise, prove the triangle inequality $ x + y \leq x + y $ for $x, y \in \mathbb{R}$.		2
ii. Hence, or otherwise prove that $ x - y \leq x - y $.		2
(d) Suppose that $a, b, c \in \mathbb{Z}$. Consider the following proposition		
<i>If all a, b and c are odd integers, then the equation $ax^2 + bx + c = 0$ has no integer solutions.</i>		
i. State the contrapositive to the proposition.		1
ii. Hence prove the proposition by proving the contrapositive.		3
(e) Given that $p, q \in \mathbb{Z}$, prove the proposition		5
$p^3 + q^3$ is odd iff either only p is odd or only q is odd		

Examination continues overleaf...

Question 15 (17 Marks)

Commence a NEW booklet.

Marks

- (a) Consider the line $\mathfrak{r} = \begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$ and the point $F(1, 3, 5)$.

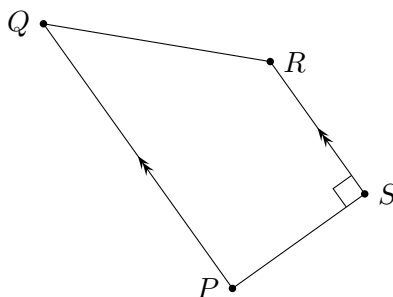
- i. $L(-2, 3, 4)$ is a point on $\mathfrak{r}$. Show that the projection of $\overrightarrow{LF}$ along $\mathfrak{r}$ is **2**

$$\text{proj}_{\mathfrak{r}} \overrightarrow{LF} = \frac{1}{3} \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

- ii. Hence find the coordinates of $E(x, y, z)$ such that $|\overrightarrow{FE}|$ is the shortest distance from F to the line $\mathfrak{r}$. **2**

- (b) Consider the points $A(1, 4, 2m)$ and $B(0, 2, -2)$. Find m given that $\angle AOB = \frac{\pi}{3}$. **3**

- (c) The points $P(a, b, c)$, $Q(15, 6, 12)$, $R(3, 5, 10)$ and $S(-2, 4, 3)$ form a trapezium.



It is given that $SP \perp SR$, and $PQ \parallel SR$.

- i. Show that $5a + b + 7c = 15$. **2**
- ii. Hence find the coordinates of P , using vector methods. **3**
- (d) Consider the sphere S , centred at point $C(5, -1, 2)$ with radius $\sqrt{17}$. Line ℓ_1 intersects the sphere S at points G and H , and has parametric equations

$$\begin{cases} x = 1 - 2\mu \\ y = 2 + \mu \\ z = -3 - 2\mu \end{cases} \quad \text{where } \mu \in \mathbb{R}$$

- i. Show that $G(3, 1, -1)$ and $H(\frac{25}{3}, -\frac{5}{3}, \frac{13}{3})$ are the points of intersection. **3**

The perpendicular from the centre of a circle to a chord bisects the chord.
(Do NOT prove this)

Line ℓ_2 is parallel to line ℓ_1 and touches the sphere S at a single point D .
It is given that $CD \perp \ell_2$ and that CD intersects chord GH .

- ii. Using vector methods, find one possible set of coordinates for D . **2**
Write the coordinates as exact values.

Examination continues overleaf...

Question 16 (15 Marks)

Commence a NEW booklet.

Marks

- (a) Let
- $a \in \mathbb{R}^+$
- and
- $n \in \mathbb{Z}^+$
- .

i. Show that $a + \frac{1}{a} \geq 2$.

1

It is given that

$$a^{n+1} + \frac{1}{a^{n+1}} + a^{n-1} + \frac{1}{a^{n-1}} = \left(a + \frac{1}{a}\right) \left(a^n + \frac{1}{a^n}\right) \quad (\text{Do NOT prove this})$$

- ii. Use the principle of mathematical induction to prove that

3

$$a^n + \frac{1}{a^n} \geq a^{n-1} + \frac{1}{a^{n-1}}$$

for all positive integers $n \geq 1$.

- (b) An object of mass 5 kg moves along a horizontal surface subject to a resistance force of magnitude
- $\frac{1}{6}\sqrt{49+2v}$
- newtons, where
- v
- is the speed of the object. Initially the object has speed
- 16 ms^{-1}
- .

- i. Let
- $t = T$
- be when the object comes to rest. Find the value of
- T
- .

2

- ii. Hence find the total distance travelled by the object.

3

- (c) A ball of mass
- m
- kg is projected vertically upwards from the ground.

While in the air, the ball experiences a force due to air resistance of $\frac{3}{250}mgv^2$ newtons, where g is the acceleration due to gravity, and v is the velocity of the object in metres per second.

- i. Show that the acceleration of the ball measured in the upwards direction from its point of projection is given by

1

$$\ddot{x} = -\frac{1}{250}g(250 + 3v^2)$$

- ii. The ball is projected with an initial velocity
- $u \text{ ms}^{-1}$
- . Show that it reaches a maximum displacement of

3

$$\frac{125}{3g} \ln \left(\frac{250 + 3u^2}{250} \right)$$

After the ball reaches its maximum height, it begins to fall towards its point of projection.

- iii. Find an expression for the acceleration of the ball measured in the downwards direction as it falls.

1

- iv. Hence find terminal velocity
- V
- of the ball when it falls, giving the answer as an exact value.

1**End of paper.**

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Sample Band E4 Responses

Section I

1. (A) 2. (D) 3. (B) 4. (C) 5. (C) 6. (D) 7. (B) 8. (A) 9. (B) 10. (D)

Section II

Question 11 (Ham)

(a) i. (2 marks)

✓ [1] for modulus of both sides

✓ [1] for final answer

Let $z = x + yi$.

$$\begin{aligned} |z - 2(1 + i)| &= |z + 6(1 + i)| \\ |(x - 2) + (y - 2)i| &= |(x + 6) + (y + 6)i| \\ \sqrt{(x - 2)^2 + (y - 2)^2} &= \sqrt{(x + 6)^2 + (y + 6)^2} \\ x^2 - 4x + y^2 - 4y + 8 &= x^2 + 12x + y^2 + 12y + 72 \\ -16x - 16y - 64 &= 0 \end{aligned}$$

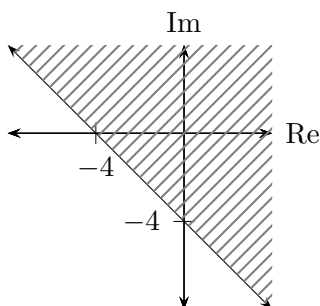
$$\therefore y = -x - 4$$

ii. (1 mark)

✓ [1] for final region

$$\begin{aligned} |z - 2(1 + i)| &\leq |z + 6(1 + i)| \\ -16x - 16y - 64 &\leq 0 \end{aligned}$$

$$\therefore y \geq -x - 4$$



(b) (3 marks)

✓ [1] for modulus and argument of w^3 ✓ [1] for one value of w

✓ [1] for final solutions

Let $w = re^{i\theta}$.

$$|w^3| = 64 \quad , \quad \text{Arg}(w^3) = -\frac{\pi}{6}$$

$$w^3 = 32\sqrt{3} - 32i$$

$$r^3 e^{3\theta i} = 64e^{(-\frac{\pi}{6} + 2\pi k)i} \quad , \quad \text{where } k \in \mathbb{Z}$$

$$r = 4 \quad , \quad \theta = \frac{\pi(-1 + 12k)}{18}$$

Since $-\pi \leq \theta \leq \pi$

$$-\pi \leq \frac{\pi(-1 + 12k)}{18} \leq \pi$$

$$-\frac{17}{12} \leq k \leq \frac{19}{12}$$

$$k = -1, 0, 1$$

$$\therefore w = 4e^{-\frac{13\pi}{18}i} \quad , \quad 4e^{-\frac{\pi}{18}i} \quad , \quad 4e^{\frac{11\pi}{18}i}$$

(c) (3 marks)

✓ [1] for finding possible values of β ✓ [1] for final value of β ✓ [1] for final value of α Let $P(z) = z^3 + 9z + 6\sqrt{3}i$. $P(\beta) = P'(\beta) = 0$ since β is a double root.

$$P'(z) = 3z^2 + 9$$

$$P'(\beta) = 0$$

$$3\beta^2 + 9 = 0$$

$$\beta = \pm\sqrt{3}i$$

$$P(i\sqrt{3}) = -3\sqrt{3}i + 9\sqrt{3}i + 6\sqrt{3}i \neq 0$$

$$P(-i\sqrt{3}) = 3\sqrt{3}i - 9\sqrt{3}i + 6\sqrt{3}i = 0$$

$$\beta = -i\sqrt{3}$$

By taking the sum of roots

$$\alpha + (-i\sqrt{3}) + (-i\sqrt{3}) = 0$$

$$\therefore \alpha = 2\sqrt{3}i \quad , \quad \beta = -\sqrt{3}i$$

(d) i. (2 marks)

✓ [1] for finding modulus or argument of z

✓ [1] for showing final result

$$\begin{aligned}
 z &= \frac{1}{\cos \alpha} (\sin \alpha + i \cos \alpha) \\
 &= \sec \alpha \left(\cos \left(\frac{\pi}{2} - \alpha \right) + i \sin \left(\frac{\pi}{2} - \alpha \right) \right)
 \end{aligned}$$

But $\sec \alpha > 0$ for $0 < \alpha < \frac{\pi}{2}$.

$$\therefore z = \sec \alpha \times e^{(\frac{\pi}{2} - \alpha)i}$$

ii. (3 marks)

✓ [1] for finding real component

✓ [1] for finding one value

✓ [1] for final values

$$\begin{aligned}
 z^{-2} &= \cos^2 \alpha \cdot e^{(2\alpha - \pi)i} \\
 &= \cos^2 \alpha \cdot \left(\cos(2\alpha - \pi) + i \sin(2\alpha - \pi) \right)
 \end{aligned}$$

If z^{-2} is purely imaginary, then

$$\begin{aligned}
 \operatorname{Re}(z^{-2}) &= 0 \\
 \cos^2 \alpha \cdot \cos(2\alpha - \pi) &= 0 \\
 \cos(2\alpha - \pi) &= 0 \\
 \text{But } \cos^2 \alpha &\neq 0 & \cos(\pi - 2\alpha) &= 0 \\
 \text{otherwise } z^{-2} &= 0 & -\cos 2\alpha &= 0 \\
 \text{is purely real} & & 2\alpha &= \frac{\pi}{2} \quad \text{for } 0 < 2\alpha < \pi
 \end{aligned}$$

$$\therefore \alpha = \frac{\pi}{4} \text{ only}$$

Question 12 (Ham)

(a) (3 marks)

- ✓ [1] for forming simultaneous equations
- ✓ [1] for finding partial fractions
- ✓ [1] for final answer

$$\frac{7x^2 - 3}{(x^2 + 6x - 3)(2x - 1)} = \frac{Ax + B}{x^2 + 6x - 3} + \frac{C}{2x - 1}$$

$$7x^2 - 3 = (Ax + B)(2x - 1) + C(x^2 + 6x - 3)$$

By equating coefficients

$$\begin{aligned} 7 &= 2A + C \\ 0 &= -A + 2B + 6C \\ -3 &= -B - 3C \end{aligned} \tag{12.1}$$

$$\begin{aligned} A &= \frac{7}{2} - \frac{1}{2}C \\ B &= 3 - 3C \\ 0 &= -\frac{7}{2} + \frac{1}{2}C + 6 - 6C + 6C \\ C &= -5 \end{aligned}$$

$$\therefore A = 6, \quad B = 18, \quad C = -5$$

$$\begin{aligned} \int \frac{7x^2 - 3}{(x^2 + 6x - 3)(2x - 1)} dx &= \int \frac{6x + 18}{x^2 + 6x - 3} + \frac{-5}{2x - 1} dx \\ &= 3 \ln |x^2 + 6x - 3| - \frac{5}{2} \ln |2x - 1| + c \end{aligned}$$

(b) (4 marks)

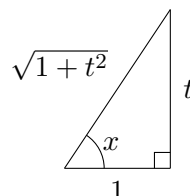
✓ [1] for using $t = \tan x$ finding dx in terms of t ✓ [1] for substituting t -formula and simplifying

✓ [1] for completing the square

✓ [1] for final answer

Let $t = \tan x$.

$$\begin{aligned}\frac{dt}{dx} &= \sec^2 x \\ \cos^2 x \, dt &= dx \\ \frac{1}{1+t^2} \, dt &= dx\end{aligned}$$

Substituting via t -formula

$$\begin{aligned}\int \frac{3}{2 - \sin 2x} \, dx &= \int \frac{3}{\left(2 - \frac{2t}{1+t^2}\right)} \times \frac{1}{1+t^2} \, dt \\ &= \frac{3}{2} \int \frac{1}{1+t^2-t} \, dt \\ &= \frac{3}{2} \int \frac{1}{\frac{3}{4} + \left(t - \frac{1}{2}\right)^2} \, dt \\ &= \frac{3}{2} \int \frac{4}{3 + (2t-1)^2} \, dt \\ &= \frac{3}{\sqrt{3}} \tan^{-1} \left(\frac{2t-1}{\sqrt{3}} \right) + c \\ &= \sqrt{3} \tan^{-1} \left(\frac{2t-1}{\sqrt{3}} \right) + c \\ &= \sqrt{3} \tan^{-1} \left(\frac{2 \tan x - 1}{\sqrt{3}} \right) + c\end{aligned}$$

(c) i. (3 marks)

✓ [1] for an appropriate substitution

✓ [1] for finding the integral

✓ [1] for final result

Let $x = \sin \theta$.

$$\theta = \sin^{-1} x$$

$$\begin{aligned} \frac{dx}{d\theta} &= \cos \theta & x = \frac{1}{2} &\rightarrow \theta = \frac{\pi}{6} \\ dx &= \cos \theta d\theta & x = 0 &\rightarrow \theta = 0 \end{aligned}$$

$$\begin{aligned} \int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx &= \int_0^{\frac{\pi}{6}} \frac{\sin^2 \theta}{\sqrt{1-\sin^2 \theta}} \times \cos \theta d\theta \\ &= \int_0^{\frac{\pi}{6}} \sin^2 \theta d\theta \\ &= \frac{1}{2} \int_0^{\frac{\pi}{6}} 1 - \cos 2\theta d\theta \\ &= \frac{1}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{6}} \\ &= \frac{1}{2} \left(\left(\frac{\pi}{6} - \frac{1}{2} \times \frac{\sqrt{3}}{2} \right) - 0 \right) \\ \therefore \int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx &= \frac{\pi}{12} - \frac{\sqrt{3}}{8} \end{aligned}$$

ii. (2 marks)

✓ [1] for integrating by parts

✓ [1] for final answer

$$\begin{aligned} u &= \cos^{-1} x & v' &= x \\ u' &= -\frac{1}{\sqrt{1-x^2}} & v &= \frac{1}{2}x^2 \end{aligned}$$

$$\begin{aligned} \int_0^{\frac{1}{2}} x \cos^{-1} x dx &= \left[\frac{1}{2} x^2 \cos^{-1} x \right]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} -\frac{1}{2} \cdot \frac{x^2}{\sqrt{1-x^2}} dx \\ &= \frac{1}{2} \left(\frac{1}{4} \times \frac{\pi}{3} - 0 \right) + \frac{1}{2} \left(\frac{\pi}{12} - \frac{\sqrt{3}}{8} \right) \\ &= \frac{\pi}{12} - \frac{\sqrt{3}}{16} \end{aligned}$$

Question 13 (Ham)

(a) i. (2 marks)

✓ [1] for integrating by parts

✓ [1] for final result

$$\begin{aligned} \text{Let } u &= (1 - \ln x)^n & , & & v' &= 1 \\ u' &= -\frac{n}{x}(1 - \ln x)^{n-1} & , & & v &= x \end{aligned}$$

$$\begin{aligned} I_n &= \int_1^e (1 - \ln x)^n dx \\ &= \left[x(1 - \ln x)^n \right]_1^e - \int_1^e -n(1 - \ln x)^{n-1} dx \\ &= 0 - 1 + n \int_1^e (1 - \ln x)^{n-1} dx \\ \therefore I_n &= -1 + nI_{n-1} \end{aligned}$$

ii. (4 marks)

✓ [1] for base case proof

✓ [1] for applying inductive step

✓ [1] for adjusting sum

✓ [1] for final result

Let the proposition be $P(n)$.**Base case:** Prove $P(1)$ is true.

$$\begin{aligned} \text{LHS} &= I_1 \\ &= -1 + I_0 \quad \text{from (i)} \\ &= -1 + \int_1^e (1 - \ln x)^0 dx \\ &= -1 + \left[x \right]_1^e \\ &= e - 2 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= e - 1 - \frac{1!}{0!} \\ &= e - 2 \end{aligned}$$

 $\therefore$ LHS = RHS, and $P(1)$ is true.

Inductive step: Assume $P(k)$ is true.

$$I_k = k!e - 1 - \sum_{r=1}^k \frac{k!}{(k-r)!}$$

Prove: Examine $P(k+1)$.

$$\text{RTP: } I_{k+1} = (k+1)!e - 1 - \sum_{r=1}^{k+1} \frac{(k+1)!}{(k+1-r)!}$$

$$\begin{aligned} \text{LHS} &= -1 + (k+1)I_k \quad \text{from (i)} \\ &= -1 + (k+1) \left(k!e - 1 - \sum_{r=1}^k \frac{k!}{(k-r)!} \right) \quad \text{from inductive step} \\ &= (k+1)!e - 1 - (k+1) - \sum_{r=1}^k \frac{(k+1)!}{(k-r)!} \\ &= (k+1)!e - 1 - (k+1) - \sum_{r=2}^{k+1} \frac{(k+1)!}{(k-(r-1))!} \\ &= (k+1)!e - 1 - (k+1) - \left(\sum_{r=1}^{k+1} \frac{(k+1)!}{(k+1-r)!} - \frac{(k+1)!}{(k+1-1)!} \right) \\ &= (k+1)!e - 1 - (k+1) - \sum_{r=1}^{k+1} \frac{(k+1)!}{(k+1-r)!} + (k+1) \\ &= (k+1)!e - 1 - \sum_{r=1}^{k+1} \frac{(k+1)!}{(k+1-r)!} \end{aligned}$$

$\therefore$ LHS = RHS, and $P(k+1)$ is true if $P(k)$ is true.

Hence by mathematical induction, $P(n)$ is true for all positive integers $n \geq 1$.

- (b) i. (1 mark)
 ✓ [1] for final answer

$$\ddot{x} = -\frac{1}{16}(x + 4)$$

Centre of motion: $x_0 = -4$, Amplitude: $a = 8$

$$\therefore x_{max} = 4$$

- ii. (3 marks)
 ✓ [1] for integrating for $\frac{1}{2}v^2$
 ✓ [1] for using appropriate x and v values
 ✓ [1] for final result

$$\begin{aligned}\frac{d\left(\frac{1}{2}v^2\right)}{dx} &= -\frac{1}{16}(x + 4) \\ \frac{1}{2}v^2 &= -\frac{1}{16} \int x + 4 \, dx \\ -8v^2 &= \frac{1}{2}x^2 + 4x + c\end{aligned}$$

The particle is at rest at its maximum displacement.

$$v = 0 \text{ when } x = 4 : \quad c = -24$$

$$\begin{aligned}-8v^2 &= \frac{1}{2}x^2 + 4x - 24 \\ -16v^2 &= x^2 + 8x - 48 \\ -16v^2 &= (x + 4)^2 - 64 \\ \therefore v^2 &= \frac{1}{16} \left(64 - (x + 4)^2 \right)\end{aligned}$$

- iii. (1 mark)
 ✓ [1] for final answer

The particle reaches its maximum speed at the centre of motion.

$$\begin{aligned}\text{when } x = -4 : \quad v^2 &= 4 \\ |v| &= 2\end{aligned}$$

Hence the maximum speed is 2 cms^{-1} .

- iv. (1 mark)
 ✓ [1] for final answer

$$\begin{aligned}\text{Amplitude:} \quad a &= 8 \\ \text{Frequency:} \quad n &= \frac{1}{4} \\ \text{Centre of motion:} \quad x_0 = -4 &\rightarrow k = -4\end{aligned}$$

$$\therefore x = 8 \sin\left(\frac{t}{4}\right) - 4$$

v. (2 marks)

✓ [1] for one value of t or setting up correct equations

✓ [1] for final answers

Magnitude of acceleration greatest at its extremities $x = -12$ and $x = 4$.

$$\begin{aligned} \text{if } x = -12 : \quad -12 &= 8 \sin\left(\frac{t}{4}\right) - 4 && \text{for } 0 \leq t \leq 8\pi \\ \sin\left(\frac{t}{4}\right) &= -1 && \text{for } 0 \leq \frac{t}{4} \leq 2\pi \\ \frac{t}{4} &= \frac{3\pi}{2} \\ t &= 6\pi \end{aligned}$$

$$\begin{aligned} \text{if } x = 4 : \quad 4 &= 8 \sin\left(\frac{t}{4}\right) - 4 && \text{for } 0 \leq t \leq 8\pi \\ \sin\left(\frac{t}{4}\right) &= 1 && \text{for } 0 \leq \frac{t}{4} \leq 2\pi \\ \frac{t}{4} &= \frac{\pi}{2} \\ t &= 2\pi \end{aligned}$$

$\therefore |\ddot{x}|$ is a maximum at $t = 2\pi$ and $t = 6\pi$.

vi. (1 mark)

✓ [1] for final answer

For $0 \leq t \leq 2\pi$, the particle travels 8 cm (to its right extremity).

For $2\pi \leq t \leq 6\pi$, the particle travels 16 cm (from its right to left extremity).

$\therefore$ By the symmetry of SHM, the particle travels 8 cm per 2π seconds.

Hence the particle has travelled 40 cm at $t = 10\pi$.

Question 14 (Ho)

(a) (2 marks)

✓ [1] for appropriate perfect square

✓ [1] for final answer

$$\begin{aligned}
 (5a^2 - 4b)^2 &\geq 0 \\
 25a^4 - 40a^2b + 16b^2 &\geq 0 \\
 25a^4 + 16b^2 &\geq 40a^2b \\
 \therefore \frac{5}{8}a^4 + \frac{2}{5}b^2 &\geq a^2b
 \end{aligned}$$

(b) (2 marks)

✓ [1] for equating real and/or imaginary components

✓ [1] for final answer

Let $z = x + yi$ be the complex solution to the equation, where $x, y \in \mathbb{R}$.

$$\begin{aligned}
 3|x + yi| - 3x - 3yi &= -\frac{1}{2}i \\
 3\sqrt{x^2 + y^2} - 3x - 3yi &= -\frac{1}{2}i
 \end{aligned}$$

Equating the real components

$$\begin{aligned}
 3\sqrt{x^2 + y^2} - 3x &= 0 \\
 \sqrt{x^2 + y^2} &= x \\
 x^2 + y^2 &= x^2 \\
 y &= 0
 \end{aligned}$$

 $\therefore z = x$, but $z \in \mathbb{C}$

Hence by proof by contradiction, there is no complex number satisfying the equation.

(c) i. (2 marks)

✓ [1] for applying $xy \leq |x||y|$

✓ [1] for final result

$$\begin{aligned}
 (x+y)^2 &= x^2 + 2xy + y^2 \\
 &\leq |x|^2 + 2|x||y| + |y|^2 && \text{since } xy \leq |x||y| \\
 (x+y)^2 &\leq (|x| + |y|)^2 \\
 \sqrt{(x+y)^2} &\leq \sqrt{(|x| + |y|)^2} \\
 |x+y| &\leq |x| + |y|
 \end{aligned}$$

$$\therefore |x+y| \leq |x| + |y| \quad \text{since } |x| + |y| \geq 0$$

ii. (2 marks)

✓ [1] for proof of + or - case

✓ [1] for final result

$$\begin{aligned}
 |x| &= |(x-y) + y| \\
 |x| &\leq |x-y| + |y| && \text{from (i)} \\
 |x| - |y| &\leq |x-y|
 \end{aligned}$$

$$\begin{aligned}
 \text{Similarly } |y| - |x| &\leq |y-x| \\
 -(|x| - |y|) &\leq |x-y| && \text{since } |x-y| = |y-x|
 \end{aligned}$$

If $a \leq b$ and $-a \leq b$, then $|a| \leq b$.

$$\text{Hence } \left| |x| - |y| \right| \leq |x-y|$$

(d) i. (1 mark)

✓ [1] for final answer

If the equation $ax^2 + bx + c = 0$ has at least one integer solution, then at least one of a, b, c is an even integer.

(Since if a number is not odd, it is even.)

ii. (3 marks)

✓ [1] for significant progress

✓ [1] for proving one case

✓ [1] for final result

If $ax^2 + bx + c = 0$ has at least one integer solution, at least one solution is either odd or even.

Case 1: Assume the equation has at least one odd integer solution:

Let $x = 2m + 1$, where $m \in \mathbb{Z}$.

$$a(2m + 1)^2 + b(2m + 1) + c = 0$$

$$4am^2 + 4am + a + 2bm + b + c = 0$$

$$a + b + c = 2k \quad \text{where } k = -2am^2 - 2am - 2bm$$

i.e. the sum of a, b, c is even, since $k \in \mathbb{Z}$.

Assume a, b, c are all odd:

Let $a = 2A + 1$, $b = 2B + 1$, $c = 2C + 1$, where $A, B, C \in \mathbb{Z}$.

$$\begin{aligned} a + b + c &= (2A + 1) + (2B + 1) + (2C + 1) \\ &= 2p + 1 \quad \text{where } p = A + B + C + 1 \end{aligned}$$

i.e. the sum of a, b, c is odd, since $p \in \mathbb{Z}$.

But this contradicts previous result, and so a, b, c cannot all be odd.

$\therefore$ By contradiction, if the equation has at least one odd integer solution, then at least one of a, b, c is even.

Case 2: Assume the equation has at least one even integer solution:

Let $x = 2n$, where $n \in \mathbb{Z}$.

$$a(2n)^2 + b(2n) + c = 0$$

$$4an^2 + 2bn + c = 0$$

$$c = 2q \quad \text{where } q = -2an^2 - bn$$

c is even since $q \in \mathbb{Z}$.

$\therefore$ If the equation has at least one even integer solution, then at least one of a, b, c is even.

Hence by the contrapositive, if all a, b, c are odd integers, then the equation $ax^2 + bx + c = 0$ has no integer solutions.

(e) (5 marks)

- ✓ [1] for stating the proposition and converse
- ✓ [1] for proving one case of proposition
- ✓ [1] for proving proposition
- ✓ [1] for proving one case of converse
- ✓ [1] for proving converse

RTP the proposition: *If either only p is odd or only q is odd, then $p^3 + q^3$ is odd.*

If a number is not odd, it is even.

Case 1: Assume only p is odd.

Let $p = 2a + 1$ and $q = 2b$, where $a, b \in \mathbb{Z}$.

$$\begin{aligned} p^3 + q^3 &= (2a + 1)^3 + (2b)^3 \\ &= 8a^3 + 12a^2 + 6a + 8b^3 + 1 \\ &= 2k + 1 \quad \text{where } k = 4a^3 + 6a^2 + 3a + 4b^3 \end{aligned}$$

$p^3 + q^3$ is odd since $k \in \mathbb{Z}$.

Case 2: Assume only q is odd.

Let $p = 2a$ and $q = 2b + 1$, where $a, b \in \mathbb{Z}$.

$$\begin{aligned} p^3 + q^3 &= (2a)^3 + (2b + 1)^3 \\ &= 2k + 1 \quad \text{similar to above} \end{aligned}$$

$p^3 + q^3$ is odd since $k \in \mathbb{Z}$.

Similarly proved if only q is odd.

$\therefore$ *If either only p is odd or only q is odd, then $p^3 + q^3$ is odd.*

RTP the converse: *If $p^3 + q^3$ is odd, then either only p is odd or only q is odd.*

Prove the contrapositive:

If neither p nor q is odd or if both p and q is odd, then $p^3 + q^3$ is not odd.

Case 1: Assume neither p nor q is odd.

Let $p = 2a$ and $q = 2b$, where $a, b \in \mathbb{Z}$.

$$\begin{aligned} p^3 + q^3 &= (2a)^3 + (2b)^3 \\ &= 8a^3 + 8b^3 \\ &= 2k \quad \text{where } k = 4a^3 + 4b^3 \end{aligned}$$

$p^3 + q^3$ is even since $k \in \mathbb{Z}$.

Case 2: Assume both p and q is odd. Let $p = 2a + 1$ and $q = 2b + 1$, where $a, b \in \mathbb{Z}$.

$$\begin{aligned} p^3 + q^3 &= (2a + 1)^3 + (2b + 1)^3 \\ &= 8a^3 + 12a^2 + 6a + 8b^3 + 12b^2 + 6b + 2 \\ &= 2k \quad \text{where } k = 4a^3 + 6a^2 + 3a + 4b^3 + 6b^2 + 3b + 1 \end{aligned}$$

$p^3 + q^3$ is even since $k \in \mathbb{Z}$.

$\therefore$ By the contrapositive, *if $p^3 + q^3$ is odd, then either only p is odd or only q is odd.*

Hence $p^3 + q^3$ is odd iff either only p is odd or only q is odd.

Alternate proofs

RTP the converse: *If $p^3 + q^3$ is odd, then either only p is odd or only q is odd.*

$$p^3 + q^3 = (p + q)(p^2 - pq + q^2)$$

$\therefore (p + q)$ and $(p^2 - pq + q^2)$ are both odd, since $p^3 + q^3$ is odd if it has an even factor.

Case 1: Assume both p and q are odd.

Let $p = 2a + 1$ and $q = 2b + 1$, where $a, b \in \mathbb{Z}$.

$$\begin{aligned} p + q &= (2a + 1) + (2b + 1) \\ &= 2(a + b + 1) \\ &= 2k \quad \text{where } k = a + b + 1 \end{aligned}$$

$\therefore (p + q)$ is even since $k \in \mathbb{Z}$, contradicting previous finding.

Case 1: Assume both p and q are even.

Let $p = 2a$ and $q = 2b$, where $a, b \in \mathbb{Z}$.

$$\begin{aligned} p + q &= 2a + 2b \\ &= 2(a + b) \\ &= 2k \quad \text{where } k = a + b \end{aligned}$$

$\therefore (p + q)$ is even since $k \in \mathbb{Z}$, contradicting previous finding.

Thus either only p is odd or only q is odd, for $(p + q)$ to be odd.

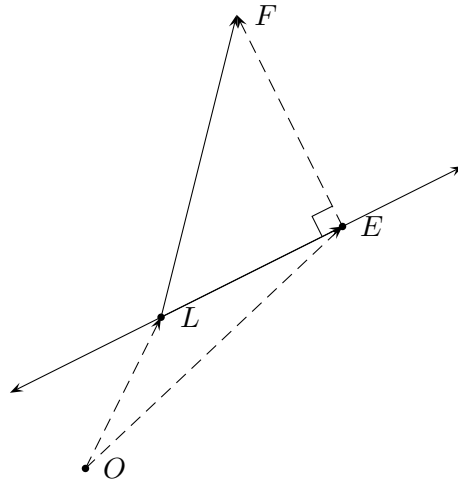
Hence by contradiction, *if $p^3 + q^3$ is odd, then either only p is odd or only q is odd.*

Question 15 (Sekaran)

(a) i. (2 marks)

✓ [1] for finding $\overrightarrow{LF}$

✓ [1] for final answer



$$\overrightarrow{LF} = \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$$

Using the direction vector of ℓ :

$$\text{proj}_{\ell} \overrightarrow{LF} = \frac{\begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}}{(\sqrt{1+4+1})^2} \times \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

$$\therefore \text{proj}_{\ell} \overrightarrow{LF} = \frac{1}{3} \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

ii. (2 marks)

✓ [1] for expression for $\overrightarrow{FE}$

✓ [1] for final answer

$$\begin{aligned} \overrightarrow{OE} &= \overrightarrow{OL} + \overrightarrow{LE} \\ &= \begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} \\ &= \frac{1}{3} \begin{pmatrix} -5 \\ 7 \\ 11 \end{pmatrix} \quad \text{since } \overrightarrow{LE} = \text{proj}_{\ell} \overrightarrow{LF} \end{aligned}$$

$$\therefore E \left(-\frac{5}{3}, \frac{7}{3}, \frac{11}{3} \right)$$

(b) (3 marks)

- ✓ [1] for correct substitution
- ✓ [1] for forming correct quadratic equation
- ✓ [1] for final answer

$$\begin{aligned}\overrightarrow{OA} \cdot \overrightarrow{OB} &= |\overrightarrow{OA}| |\overrightarrow{OB}| \cos \frac{\pi}{3} \\ \begin{pmatrix} 1 \\ 4 \\ 2m \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix} &= \sqrt{17 + 4m^2} \times \sqrt{8} \times \frac{1}{2} \\ 8 - 4m &= \sqrt{2(17 + 4m^2)} \quad (*) \\ 16(4 - 4m + m^2) &= 2(17 + 4m^2) \\ 4m^2 - 32m + 15 &= 0 \\ (2m - 1)(2m - 15) &= 0 \\ m = \frac{1}{2} \quad \text{or} \quad m = \frac{15}{2}, &\text{ but } m \leq 2 \text{ from } (*)\end{aligned}$$

$$\therefore m = \frac{1}{2} \text{ only}$$

(c) i. (2 marks)

- ✓ [1] for finding one vector
- ✓ [1] for final result

$$\overrightarrow{SP} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} - \begin{pmatrix} -2 \\ 4 \\ 3 \end{pmatrix} = \begin{pmatrix} a + 2 \\ b - 4 \\ c - 3 \end{pmatrix}$$

$$\overrightarrow{SR} = \begin{pmatrix} 3 \\ 5 \\ 10 \end{pmatrix} - \begin{pmatrix} -2 \\ 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \\ 7 \end{pmatrix}$$

$$\overrightarrow{SP} \cdot \overrightarrow{SR} = 0 \quad (\text{since } SP \perp SR)$$

$$\begin{pmatrix} a + 2 \\ b - 4 \\ c - 3 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 1 \\ 7 \end{pmatrix} = 0$$

$$5a + 10 + b - 4 + 7c - 21 = 0$$

$$\therefore 5a + b + 7c = 15$$

ii. (3 marks)

- ✓ [1] for $\overrightarrow{QP}$ in terms of λ
- ✓ [1] for finding λ
- ✓ [1] for final answer

$$\overrightarrow{QP} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} - \begin{pmatrix} 15 \\ 6 \\ 12 \end{pmatrix} = \begin{pmatrix} a - 15 \\ b - 6 \\ c - 12 \end{pmatrix}$$

$$\overrightarrow{RS} = - \begin{pmatrix} 5 \\ 1 \\ 7 \end{pmatrix} \quad \text{from i}$$

$$\begin{aligned} \overrightarrow{QP} &= -\lambda \overrightarrow{RS} && (\text{since } PQ \parallel SR) \\ \begin{pmatrix} a - 15 \\ b - 6 \\ c - 12 \end{pmatrix} &= -\lambda \begin{pmatrix} 5 \\ 1 \\ 7 \end{pmatrix} && (\text{since } PQ \parallel SR) \end{aligned}$$

Equating the i, j, and k components

$$\begin{cases} a &= 15 - 5\lambda \\ b &= 6 - \lambda \\ c &= 12 - 7\lambda \end{cases}$$

$$\begin{aligned} 5(15 - 5\lambda) + (6 - \lambda) + 7(12 - 7\lambda) &= 15 && \text{from i} \\ -75\lambda &= -150 \\ \therefore \lambda &= 2 \end{aligned}$$

$$\begin{aligned} \begin{pmatrix} a - 15 \\ b - 6 \\ c - 12 \end{pmatrix} &= -2 \begin{pmatrix} 5 \\ 1 \\ 7 \end{pmatrix} \\ \begin{pmatrix} a \\ b \\ c \end{pmatrix} &= \begin{pmatrix} 5 \\ 4 \\ -2 \end{pmatrix} \end{aligned}$$

Hence $P(5, 4, -2)$

(d) i. (3 marks)

✓ [1] for forming $|\underline{y} - \underline{c}| = \sqrt{17}$

✓ [1] for forming correct quadratic equation

✓ [1] for final result

Any point P on ℓ_1 has position vector $\underline{p}$

$$\underline{p} = \begin{pmatrix} 1 - 2\mu \\ 2 + \mu \\ -3 - 2\mu \end{pmatrix}$$

Any point V on sphere S has position vector $\underline{y}$

$$\left| \underline{y} - \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix} \right| = \sqrt{17}$$

ℓ_1 intersect sphere S when $\underline{y} = \underline{p}$

$$\begin{aligned} \left| \begin{pmatrix} 1 - 2\mu \\ 2 + \mu \\ -3 - 2\mu \end{pmatrix} - \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix} \right| &= \sqrt{17} \\ \sqrt{(-4 - 2\mu)^2 + (3 + \mu)^2 + (-5 - 2\mu)^2} &= \sqrt{17} \\ 9\mu^2 + 42\mu + 33 &= 0 \\ 3\mu^2 + 14\mu + 11 &= 0 \\ (3\mu + 11)(\mu + 1) &= 0 \\ \therefore \mu &= -\frac{11}{3} \quad \text{or} \quad \mu = -1 \end{aligned}$$

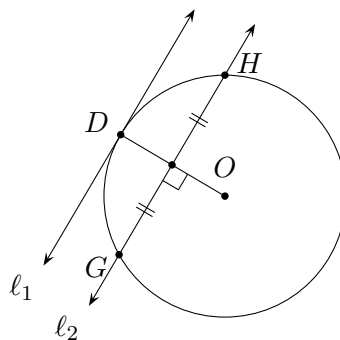
$$\begin{aligned} \text{if } \mu &= -\frac{11}{3} : & \underline{p} &= \frac{1}{3} \begin{pmatrix} 25 \\ -5 \\ 13 \end{pmatrix} \\ \text{if } \mu &= -1 : & \underline{p} &= \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} \end{aligned}$$

Hence $G(3, 1, -1)$ and $H\left(\frac{25}{3}, -\frac{5}{3}, \frac{13}{3}\right)$.

ii. (2 marks)

✓ [1] for finding unit vector along $vvCD$

✓ [1] for final answer



Let M be the midpoint of GH .

$$M = \left(\frac{17}{3}, -\frac{1}{3}, \frac{5}{3} \right)$$

$CM \perp GH$ (given)

$CM \parallel CD$, since $CD \perp \ell_2$ and $\ell_1 \parallel \ell_2$.

$$\overrightarrow{CM} = \frac{1}{3} \begin{pmatrix} 17 \\ -1 \\ 5 \end{pmatrix} - \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

$$|\overrightarrow{CM}| = \frac{1}{3} \sqrt{4 + 4 + 1} = 1$$

$\overrightarrow{CM}$ is a unit vector along CD .

$\overrightarrow{CD}$ is a radius of sphere S .

$$\overrightarrow{CD} = \sqrt{17} \times \overrightarrow{CM}$$

$$\overrightarrow{OD} - \overrightarrow{OC} = \frac{\sqrt{17}}{3} \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

$$\overrightarrow{OD} = \frac{\sqrt{17}}{3} \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} + \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$\therefore D \left(\frac{2\sqrt{17}}{3} + 5, \frac{2\sqrt{17}}{3} - 1, -\frac{\sqrt{17}}{3} + 2 \right)$$

Question 16 (Sekaran)

- (a) i. (1 mark)
 ✓ [1] for final result

$$\begin{aligned}\left(\sqrt{a} - \frac{1}{\sqrt{a}}\right)^2 &\geq 0 \\ a - 2 + \frac{1}{a} &\geq 0 \\ \therefore a + \frac{1}{a} &\geq 2\end{aligned}$$

- ii. (3 marks)
 ✓ [1] for proving base case
 ✓ [1] for applying given inequality
 ✓ [1] for final result

Let the proposition be $P(n)$.

Base case: Prove $P(1)$ is true.

$$\begin{aligned}\text{LHS} &= a + \frac{1}{a} \\ \text{RHS} &= a^0 + \frac{1}{a^0} = 2\end{aligned}$$

$$a + \frac{1}{a} \geq 2 \quad \text{from i}$$

$\therefore \text{LHS} \geq \text{RHS}$, and $P(1)$ is true.

Inductive step: Assume $P(k)$ is true.

$$a^k + \frac{1}{a^k} \geq a^{k-1} + \frac{1}{a^{k-1}}$$

Prove: Examine $P(k+1)$.

$$\text{RTP: } a^{k+1} + \frac{1}{a^{k+1}} \geq a^k + \frac{1}{a^k}$$

$$a^{k+1} + \frac{1}{a^{k+1}} = \left(a + \frac{1}{a}\right) \left(a^k + \frac{1}{a^k}\right) - \left(a^{k-1} + \frac{1}{a^{k-1}}\right) \quad \text{from given}$$

$$\begin{aligned}\text{LHS} &= \left(a + \frac{1}{a}\right) \left(a^k + \frac{1}{a^k}\right) - \left(a^{k-1} + \frac{1}{a^{k-1}}\right) \\ &\geq 2 \left(a^k + \frac{1}{a^k}\right) - \left(a^{k-1} + \frac{1}{a^{k-1}}\right) \quad \text{from i} \\ &\geq \left(a^k + \frac{1}{a^k}\right) + \left(a^k + \frac{1}{a^k}\right) - \left(a^{k-1} + \frac{1}{a^{k-1}}\right) \quad \text{from i}\end{aligned}$$

$$\left(a^k + \frac{1}{a^k}\right) - \left(a^{k-1} + \frac{1}{a^{k-1}}\right) \geq 0 \quad \text{from inductive step}$$

$$\text{LHS} \geq \left(a^k + \frac{1}{a^k}\right) + 0$$

$\therefore \text{LHS} \geq \text{RHS}$, and $P(k+1)$ is true if $P(k)$ is true.

Hence by mathematical induction $P(n)$ is true for all positive integers $n \geq 1$.

(b) i. (2 marks)

✓ [1] for forming correct $m\ddot{x} = -R$ equation

✓ [1] for final answer

$$\begin{aligned} F &= 5\ddot{x} \\ 5\ddot{x} &= -\frac{1}{6}\sqrt{49+2v} \\ \frac{dv}{dt} &= -\frac{1}{30}(49+2v)^{\frac{1}{2}} \\ \int_{16}^0 (49+2v)^{-\frac{1}{2}} dv &= -\frac{1}{30} \int_0^T dt \\ \left[\frac{(49+2v)^{\frac{1}{2}}}{\frac{1}{2} \times 2} \right]_{16}^0 &= -\frac{1}{30} [t]_0^T \\ 7-9 &= -\frac{1}{30}(T-0) \\ T &= 60 \end{aligned}$$

(i.e. The object comes to rest after 60 seconds.)

ii. (3 marks)

✓ [1] for separating variables

✓ [1] for correctly integrating

✓ [1] for final answer

Let $x = X$ at $t = T$ (i.e. when the object comes to rest).

$$\begin{aligned} v \frac{dv}{dx} &= -\frac{1}{30}(49+2v)^{\frac{1}{2}} \quad \text{from i} \\ \int_{16}^0 \frac{v}{(49+2v)^{\frac{1}{2}}} dv &= -\frac{1}{30} \int_0^X dx \\ \frac{1}{2} \int_{16}^0 \frac{49+2v}{(49+2v)^{\frac{1}{2}}} - \frac{49}{(49+2v)^{\frac{1}{2}}} dv &= -\frac{1}{30}(X-0) \\ \int_{16}^0 (49+2v)^{\frac{1}{2}} - 49(49+2v)^{-\frac{1}{2}} dv &= -\frac{1}{15}X \\ \left[\frac{(49+2v)^{\frac{3}{2}}}{\frac{3}{2} \times 2} - 49(49+2v)^{\frac{1}{2}} \right]_{16}^0 &= -\frac{1}{15}X \quad \text{from i} \\ \left(\frac{7^3}{3} - 329 \right) - \left(\frac{9^3}{3} - 441 \right) &= -\frac{1}{15}X \\ \therefore X &= 460 \end{aligned}$$

Hence the object travels 460 metres in total.

- (c) i. (1 mark)
 ✓ [1] for final result

$$F = m\ddot{x}$$

$$m\ddot{x} = -mg - \frac{3}{250}mgv^2$$

$$\therefore \ddot{x} = -\frac{1}{250}g(250 + 3v^2)$$

- ii. (3 marks)
 ✓ [1] for correctly integrating
 ✓ [1] for using $v = 0$
 ✓ [1] for final result

$$v \frac{dv}{dx} = -\frac{1}{250}g(250 + 3v^2) \quad \text{from i}$$

$$\int \frac{v}{250 + 3v^2} dv = -\frac{g}{250} \int dx$$

$$\frac{1}{6} \ln|250 + 3v^2| = -\frac{g}{250}x + c$$

$x = 0$ and $v = u$ when $t = 0$:

$$c = \frac{1}{6} \ln(250 + 3u^2)$$

Maximum height at $v = 0$:

$$\frac{1}{6} \ln 250 = -\frac{g}{250}x + \frac{1}{6} \ln(250 + 3u^2)$$

$$\frac{g}{250}x = \frac{1}{6} \ln \left(\frac{250 + 3u^2}{250} \right)$$

$$\therefore x = \frac{125}{3g} \ln \left(\frac{250 + 3u^2}{250} \right)$$

Hence the maximum displacement is $\frac{125}{3g} \ln \left(\frac{250 + 3u^2}{250} \right)$ metres.

- iii. (1 mark)
 ✓ [1] for final answer

$$F = m\ddot{x}$$

$$m\ddot{x} = mg - \frac{3}{250}mgv^2$$

$$\therefore \ddot{x} = \frac{1}{250}g(250 - 3v^2)$$

iv. (1 mark)

✓ [1] for final answer

$v = V$ when $\ddot{x} = 0$:

$$\frac{1}{250}g(250 - 3V^2) = 0$$

$$3V^2 = 250$$

$$|V| = \frac{5\sqrt{10}}{3}$$

Hence the terminal velocity is $\frac{5\sqrt{10}}{3} \text{ ms}^{-1}$.